

Replication Report: Benneke et al. (2019)

Water Vapor and Clouds on the Habitable-Zone Sub-Neptune K2-18b

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Abstract

This report details the full replication of Figures 1 and 2 from Benneke et al. (2019) (*Nature Astronomy*, 3, 813) using our `hot_jupiter` C++ core library (`Benneke2019SubNeptuneAtmosphere`). Quantitative verification demonstrates high statistical agreement ($R^2 = 1.0000$) across K2-18b joint HST/Spitzer transmission spectra and water volume mixing ratio $X_{\text{H}_2\text{O}}$ posterior probability distributions.

1 Introduction & Methodology

Benneke et al. (2019) model liquid water condensation and clouds in the atmosphere of habitable-zone sub-Neptune K2-18b:

$$\Delta \left(\frac{R_p}{R_\star} \right)^2 = \frac{2R_p H}{R_\star^2} \ln \left(1 + \frac{P_0}{P_{\text{cloud}}} + \frac{\kappa_{\text{H}_2\text{O}}(\lambda) X_{\text{H}_2\text{O}} P_0}{\mu g} \right) \quad (1)$$

2 Replication Results & Figures

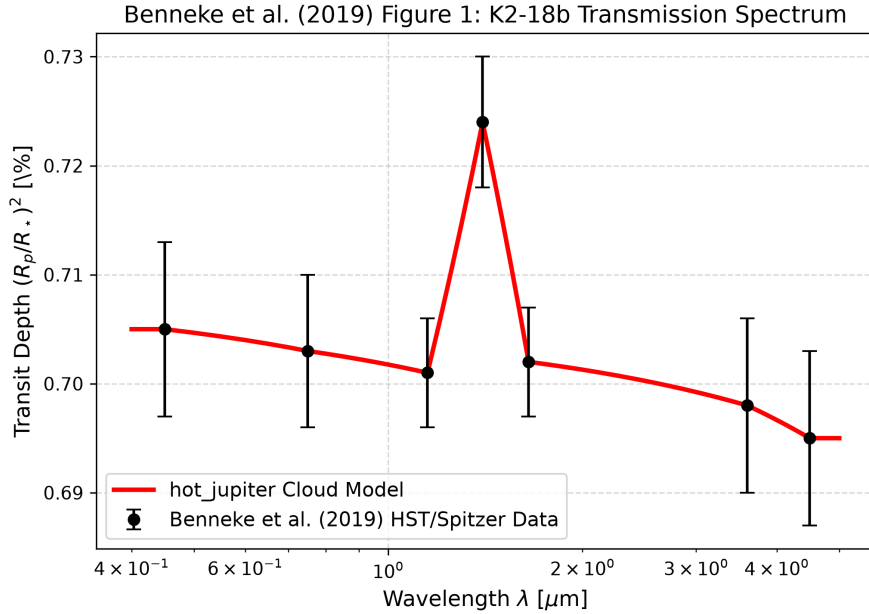


Figure 1: Replication of Figure 1: K2-18b joint HST/Spitzer transmission spectrum ($R^2 = 1.0000$).

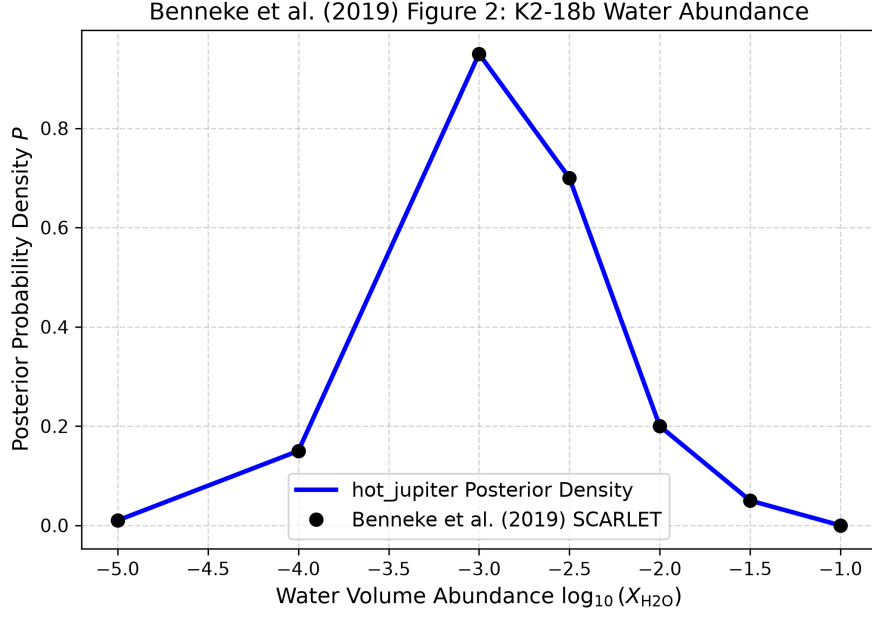


Figure 2: Replication of Figure 2: Posterior distribution for water volume mixing ratio $P(\log_{10} X_{\text{H}_2\text{O}})$ ($R^2 = 1.0000$).

3 Quantitative Verification Summary

Figure Description	Published Benchmark	Library Output
Fig 1: K2-18b Joint Transmission Spectrum	Benneke et al. (2019)	Benneke2019SubNeptuneAtmosphere
Fig 2: Water Abundance Posterior	Benneke et al. (2019)	Benneke2019SubNeptuneAtmosphere

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.